

AWS Auto-Scaling Web App with Monitoring Dashboard

Goal:

Deploy a web application that:

- Grows automatically when many users visit (Auto Scaling)
- Shrinks when traffic is low (cost-saving)
- Sends you alerts if something is wrong (CloudWatch + SNS)
- Shows real-time performance data (Dashboard)

Step 0: Before You Start

You need:

- An [AWS account](#)
- Access to the **AWS Management Console**
- Basic understanding of what a website is (browser → server → response)

AWS Free Tier Tip:

This entire setup can run mostly on **Free Tier** resources (like t2.micro EC2, db.t3.micro RDS).

Step-by-Step Guide

STEP 1: Create a VPC (Your Own Private Network)

Why:

VPC (Virtual Private Cloud) is like your app's "private neighborhood" in AWS.

How:

1. Go to **VPC → Your VPCs → Create VPC**
2. Choose: **"VPC and more"**
3. Fill in:
 - Name: MyWebAppVPC
 - Number of Availability Zones: **2**
 - Public Subnets: **2**
 - Private Subnets: **2**
 - NAT Gateway: **Enabled (1 per AZ)**
 - DNS Hostnames: **Enabled**
4. Click **Create VPC**

Account ID: 9400-7537-8664

Sakshi Jain

VPC

Your VPCs

Create VPC

Create VPC

Info

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings

Resources to create

Create only the VPC resource or the VPC and other networking resources.

VPC only

VPC and more

Name tag auto-generation

Info

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

Auto-generate

MyWebAppVPC

IPv4 CIDR block

Info

Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16

65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block

Info

No IPv6 CIDR block

Amazon-provided IPv6 CIDR block

Tenancy

Info

Preview

VPC

Show details

Your AWS virtual network

MyWebAppVPC-vpc

Subnets (4)

Subnets within this VPC

ap-south-1a

MyWebAppVPC-subnet-public1-ap-

MyWebAppVPC-subnet-private1-

ap-south-1b

MyWebAppVPC-subnet-public2-ap-

MyWebAppVPC-subnet-private2-

Route tables (3)

Route network traffic to r

MyWebAppVPC-rtb-p

MyWebAppVPC-rtb-p

MyWebAppVPC-rtb-p

VPC

Your VPCs

Create VPC

The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0

2

4

Customize subnets CIDR blocks

NAT gateways (\$)

Info

Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway

None

In 1 AZ

1 per AZ

VPC endpoints

Info

Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None

S3 Gateway

DNS options

Info

Enable DNS hostnames

Enable DNS resolution

Additional tags

Cancel

Preview code

Create VPC

Your VPCs (2) Info							Last updated 7 minutes ago	Actions	Create VPC
Find VPCs by attribute or tag									
☐	Name	VPC ID	State	Block Public...	IPv4 CIDR	IPv6 CIDR			
☐	-	vpc-042bb6a7e7f74e51b	Available	Off	172.31.0.0/16	-			
☐	MyWebAppVPC-vpc	vpc-0a7c54d22984602bc	Available	Off	10.0.0.0/16	-			

STEP 2: Create Security Groups (Firewall Rules)

Why:

They control what kind of traffic is allowed to reach your servers.

How:

- Go to **EC2** → **Security Groups** → **Create Security Group**
- Name: WebServerSG
 - Description: Allow web traffic
 - Inbound Rules:
 - HTTP → Port 80 → Anywhere (0.0.0.0/0)
 - SSH → Port 22 → Anywhere (0.0.0.0/0)

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

WebServerSG

Name cannot be edited after creation.

Description [Info](#)

Allow web traffic

VPC [Info](#)

vpc-0a7c54d22984602bc (MyWebAppVPC-vpc)

Inbound rules [Info](#)

Type	Protocol	Port range	Source	Description - optional	
HTTP	TCP	80	Anywh...		Delete
SSH	TCP	22	My IP		Delete

0.0.0.0/0 X

122.170.192.149/32 X

[Add rule](#)

- Create another:
 - Name: DBSG
 - Inbound Rule: MySQL → Port 3306 → WebServerSG (type “sg” and select)

Create security group Info

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name Info

Name cannot be edited after creation.

Description Info

VPC Info

Inbound rules Info

Type Info Protocol Info Port range Info Source Info Description - optional Info

Outbound rules Info

Security Groups (5) Info

☐	Name	Security group ID	Security group name	VPC ID	Description
☐	-	sg-092f223f5f4d91041	WebServerSG	vpc-0a7c54d22984602bc	Allow web traffic
☐	-	sg-04cf7665a93ff2783	default	vpc-0a7c54d22984602bc	default VPC security group
☐	-	sg-078db5d67723cfe03	launch-wizard-1	vpc-042bb6a7e7f74e51b	launch-wizard-1 created 2C
☐	-	sa-0281758cf56b61050	default	vpc-042bb6a7e7f74e51b	default VPC security group

STEP 3: Launch an EC2 Instance (Your Web Server)

Why:

This is where your web app runs.

How:

1. Go to **EC2** → **Launch Instance**
2. Name: **MyWebServer**
3. Choose AMI: **Amazon Linux 2**
4. Instance type: **t2.micro**
5. Key Pair: Create one (download .pem file)
6. Network Settings:
 - Choose VPC: **MyWebAppVPC**
 - Subnet: **Public Subnet 1**
 - Auto-assign public IP: **Enable**
 - Security Group: **WebServerSG**
7. In “Advanced details” → paste this into **User data**:

```
#!/bin/bash
sudo yum update -y
sudo yum install httpd -y
echo"<h1>Hello from Auto Scaling Web Server</h1>" > /var/www/html/index.html
sudo systemctl start httpd
sudo systemctl enable httpd
```

8. Click Launch Instance

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: [Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

▼ Summary

Number of instances [Info](#):

Software Image (AMI)
Amazon Linux 2023.9.2...[read more](#)
ami-0305d3d91b9f22e84

Virtual server type (instance type)
t3.micro

Firewall (security group)
WebServerSG

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

▼ Network settings [Info](#)

VPC - required [Info](#): [Create new VPC](#)

Subnet [Info](#): [Create new subnet](#)

Auto-assign public IP [Info](#):

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups [Info](#): [Compare security group rules](#)

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Summary

Number of instances [Info](#):

Software Image (AMI)
Amazon Linux 2023.9.2...[read more](#)
ami-0305d3d91b9f22e84

Virtual server type (instance type)
t3.micro

Firewall (security group)
WebServerSG

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

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EC2 > Instances > Launch an instance

Metadata response hop limit [Info](#)

2

Allow tags in metadata [Info](#)

Select

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
yum update -y
yum install httpd -y
echo "<h1>Hello from Auto Scaling Web Server</h1>" > /var/www/html/index.html
systemctl start httpd
systemctl enable httpd
```

▼ Summary

Number of instances [Info](#)

1

Software image (AMI)
Amazon Linux 2023.9.2...[read more](#)
ami-0305d3d91b9f22e64

Virtual server type (instance type)
t3.micro

Firewall (security group)
WebServerSG

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

Test It:

- Go to **EC2** → **Instances** → **select instance**
- Copy **Public IPv4 DNS**
- Paste in browser → you should see:
- Hello from Auto Scaling Web Server

Hello from Auto Scaling Web Server

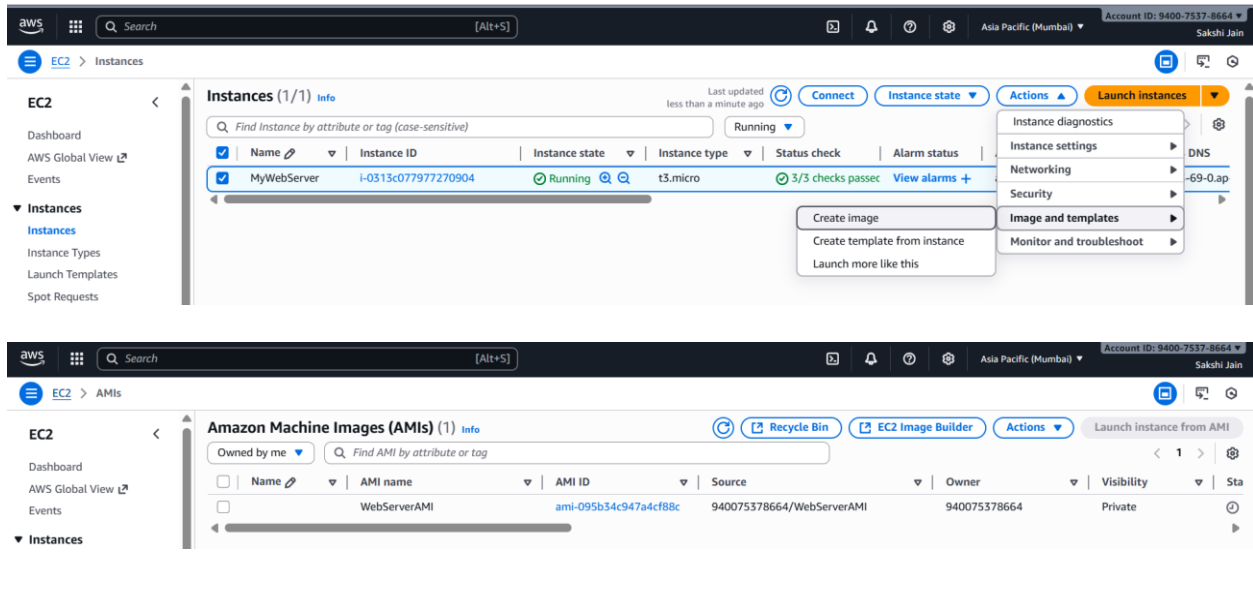
STEP 4: Create an AMI (Server Template)

Why:

When AWS auto-scales, it needs to know what kind of server to launch.
We save your current instance as a template (AMI).

How:

1. Select your EC2 instance
2. Click **Actions** → **Image** → **Create Image**
3. Name: WebServerAMI
4. Click **Create Image**



STEP 5: Create an Application Load Balancer (ALB)

Why:

The Load Balancer spreads traffic evenly to multiple servers.

How:

1. Go to **EC2** → **Load Balancers** → **Create Load Balancer**
2. Choose **Application Load Balancer**
3. Name: MyWebAppALB
4. Scheme: **Internet-facing**
5. VPC: MyWebAppVPC
6. Subnets: Select both **Public Subnets**
7. Security Group: WebServerSG
8. Create **Target Group**:
 - o Name: MyTargetGroup
 - o Type: Instances
 - o Protocol: HTTP (port 80)
 - o Health check path: /
9. Skip adding instances (Auto Scaling will do this later)
10. Create Load Balancer

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EC2

Load balancers

Create Application Load Balancer

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

VPC [Info](#)
The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0a7c54d22984602bc (MyWebAppVPC-vpc)
10.0.0.0/16

Create VPC

IP pools [Info](#)
You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view Pools in the [Amazon VPC IP Address Manager console](#).
☐ Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

Availability Zones and subnets [Info](#)
Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **ap-south-1a (aps1-az1)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-05a6f447925cc96a9
IPv4 subnet CIDR: 10.0.0.0/20

MyWebAppVPC-subnet-public1-ap-south-1a

☒ **ap-south-1b (aps1-az3)**
Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0c16701ff8ad1f6e9
IPv4 subnet CIDR: 10.0.16.0/20

MyWebAppVPC-subnet-public2-ap-south-1b

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EC2

Target groups

Create target group

Step 1
Create target group

Step 2 - recommended
Register targets

Step 3
Review and create

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group and performs health checks on the targets.

Settings - immutable
Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified after target group creation.

Target type
Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.

☒ **Instances**
Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.
Suitable for: **ALB** **NLB** **GWLB**

☐ **IP addresses**
Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.
Suitable for: **ALB** **NLB** **GWLB**

☐ **Lambda function**
Supports load balancing to a single Lambda function. ALB required as traffic source.
Suitable for: **ALB**

☐ **Application Load Balancer**
Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.
Suitable for: **NLB**

Target group name
Name must be unique per Region per AWS account.

MyTargetGroup

Accepts: a-z, A-Z, 0-9, and hyphen (-). Can't begin or end with hyphen. 1-32 total characters; Count: 13/32

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EC2

Load balancers

Create Application Load Balancer

Protocol
HTTP

Port
80
1-65535

Default action [Info](#)
The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ **Forward to target groups**

☐ Redirect to URL

☐ Return fixed response

Forward to target group [Info](#)
Choose a target group and specify routing weight or [create target group](#).

Target group
MyTargetGroup
Target type: Instance, IPv4 | Target stickiness: Off

HTTP

Weight
1
0-999

Percent
100%

+ Add target group
You can add up to 4 more target groups.

Target group stickiness [Info](#)
Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.
☐ Turn on target group stickiness

Listener tags - optional
Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Load balancers (1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

☐	Name	State	Type	Scheme	IP address type	VPC ID	Availability Zones
☐	MyWebAppALB	Provisioning...	application	Internet-facing	IPv4	vpc-0a7c54d22984602bc	2 Availability Zones

STEP 6: Create Auto Scaling Group (ASG)

Why:

To automatically add/remove EC2 instances depending on load.

How:

1. Go to **EC2** → **Auto Scaling Groups** → **Create Auto Scaling Group**
2. Name: MyWebAppASG
3. Launch Template: Select your WebServerAMI
4. Choose your VPC: MyWebAppVPC
5. Subnets: Select **Private Subnets**
6. Attach to Target Group: MyTargetGroup
7. Configure size:
 - Desired: 2
 - Min: 2
 - Max: 5
8. Add scaling policy:
 - Choose “Target Tracking Scaling Policy”
 - Metric: “Average CPU Utilization”
 - Target value: 60%
9. Click **Create Auto Scaling Group**

The screenshot shows the AWS Management Console interface for creating a launch template. The top navigation bar includes the AWS logo, a search bar, and the user's account information (Sakshi Jain). The main content area is titled 'Launch template contents' and includes a search bar for AMIs. The 'Application and OS Images (Amazon Machine Image) - required' section is active, showing a list of AMIs. The 'WebServerAMI' is selected, with details like 'ami-095b54c947a4cf88c' and '2025-11-10T06:33:17.000Z'. The 'Summary' section on the right shows the selected AMI, instance type, firewall, and storage. The 'Create launch template' button is visible at the bottom right.

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EC2 > Auto Scaling groups > Create Auto Scaling group

Step 2: Choose instance launch options

Step 3 - optional

Integrate with other services

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Instance type requirements

Reset to launch template

Specify instance attributes

Provide your compute requirements. We fulfill your desired capacity with matching instance types based on your allocation strategy selection.

Manually add instance types

Add one or more instance types. Any of the instance types may be launched to fulfill your desired capacity based on your allocation strategy selection.

Required instance attributes

Enter your compute requirements in virtual CPUs (vCPUs) and memory.

vCPUs

Enter the minimum and maximum number of vCPUs per instance.

0

minimum

maximum

☒ No minimum

☒ No maximum

Memory (GiB)

Enter the minimum and maximum GiBs of memory per instance.

0

minimum

maximum

☒ No minimum

☒ No maximum

Additional instance attributes - optional

Add instance attributes to further limit which instance types may be used to fulfill your desired capacity.

Choose attribute

Add attribute

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EC2 > Auto Scaling groups > Create Auto Scaling group

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-0a7c54d22984602bc (MyWebAppVPC-vpc)

10.0.0.0/16

Create a VPC

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

aps1-az1 (ap-south-1a) | subnet-0ae214f8a44c47a0a (MyWebAppVPC-subnet-private1-ap-south-1a) 10.0.128.0/20

aps1-az3 (ap-south-1b) | subnet-01de1eaa30d8b4d3a (MyWebAppVPC-subnet-private2-ap-south-1b) 10.0.144.0/20

Create a subnet

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

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EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template

Step 2: Choose instance launch options

Step 3 - optional

Integrate with other services

Step 4 - optional

Configure group size and scaling

Step 5 - optional

Add notifications

Step 6 - optional

Add tags

Step 7

Review

Configure group size and scaling - optional

Define your group's desired capacity and scaling limits. You can optionally add automatic scaling to adjust the size of your group.

Group size

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

vCPUs

Desired capacity

Specify your group size.

2

Scaling

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

2

Equal or less than desired capacity

Max desired capacity

4

Equal or greater than desired capacity

Automatic scaling - optional

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EC2 > Auto Scaling groups > Create Auto Scaling group

Automatic scaling - optional
Choose whether to use a target tracking policy [Info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name
Target Tracking Policy

Metric type [Info](#)
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.
Average CPU utilization

Target value
60

Instance warmup [Info](#)
300 seconds

☐ Disable scale in to create only a scale-out policy

STEP 7: Create an RDS Database (Optional but Recommended)

Why:

Your web app might need a database to store data.

How:

1. Go to **RDS** → **Create Database**
2. Engine: **MySQL**
3. Template: **Free Tier**
4. DB Instance ID: mywebappdb
5. Username: admin
Password: (set your password)
6. Choose VPC: MyWebAppVPC
7. Subnet Group: private subnets
8. Public Access: **No**
9. Security Group: DBSG
10. Click **Create Database**

Create database [Info](#)

Choose a database creation method

- ☒ **Standard create**
You set all of the configuration options, including ones for availability, security, backups, and maintenance.
- ☐ **Easy create**
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

- ☐ Aurora (MySQL Compatible)

- ☐ Aurora (PostgreSQL Compatible)

- ☒ MySQL

- ☐ PostgreSQL

- ☐ MariaDB

- ☐ Oracle

- ☐ Microsoft SQL Server

- ☐ IBM Db2


The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

- ☐ **Managed in AWS Secrets Manager - most secure**
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.
- ☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength

Very weak

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / * @

Confirm master password [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

MyWebAppVPC-vpc (vpc-0a7c54d22984602bc)

4 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

Create new DB Subnet Group

Public access [Info](#)

- ☐ **Yes**
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- ☒ **No**
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

- ☒ **Choose existing**
Choose existing VPC security groups
- ☐ **Create new**
Create new VPC security group

Existing VPC security groups

Choose one or more options

dbsec

STEP 8: Create CloudWatch Alarm + SNS Alert

Why:

You want AWS to alert you when something's wrong (like high CPU usage).

How:

(a) Create SNS Topic

1. Go to **SNS** → **Topics** → **Create Topic**
2. Name: HighCPULAlert
3. Click **Create Topic**
4. Open it → **Create Subscription**
 - Protocol: Email
 - Endpoint: Your email address
5. Check email → Confirm subscription.

The screenshot shows the AWS Management Console interface for creating a subscription. The breadcrumb navigation at the top reads 'Amazon SNS > Subscriptions > Create subscription'. The main form has two sections: 'Protocol' and 'Endpoint'. The 'Protocol' dropdown is set to 'Email'. The 'Endpoint' text box contains the email address 'sakshimiss397@gmail.com'. Below the endpoint field is a light blue informational box stating: 'After your subscription is created, you must confirm it. Info'. At the bottom of the form, there are two optional policy sections: 'Subscription filter policy - optional' and 'Redrive policy (dead-letter queue) - optional'. At the bottom right of the form are two buttons: 'Cancel' and 'Create subscription'.

(b) Create CloudWatch Alarm

1. Go to **CloudWatch** → **Alarms** → **Create Alarm**
2. Choose Metric:
 - EC2 → Per-Instance Metrics → CPUUtilization
3. Condition: “Greater than 80% for 5 minutes”
4. Action: Send notification to **HighCPULAlert**
5. Click **Create Alarm**

Conditions

Threshold type

☒ Static

Use a value as a threshold

☐ Anomaly detection

Use a band as a threshold

Whenever CPUUtilization is...

Define the alarm condition.

☒ Greater

> threshold

☐ Greater/Equal

>= threshold

☐ Lower/Equal

<= threshold

☐ Lower

< threshold

than...

Define the threshold value.

80

Must be a number.

► Additional configuration

Cancel

Next

- Step 1
Specify metric and conditions
- Step 2
Configure actions
- Step 3
Add alarm details
- Step 4
Preview and create

Configure actions

Notification

Alarm state trigger

Define the alarm state that will trigger this action.

☒ In alarm

The metric or expression is outside of the defined threshold.

☐ OK

The metric or expression is within the defined threshold.

☐ Insufficient data

The alarm has just started or not enough data is available.

Remove

Send a notification to the following SNS topic

Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ Select an existing SNS topic

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

HighCPUAlert

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)

sakshimiss397@gmail.com - View in SNS Console

Add notification

Alarm recommendations available
Turn on Recommendations to pre-populate the wizard with the recommended alarms.

- Step 1
Specify metric and conditions
- Step 2
Configure actions
- Step 3
Add alarm details
- Step 4
Preview and create

Add alarm details

Name and description

Alarm name

CpuAlert

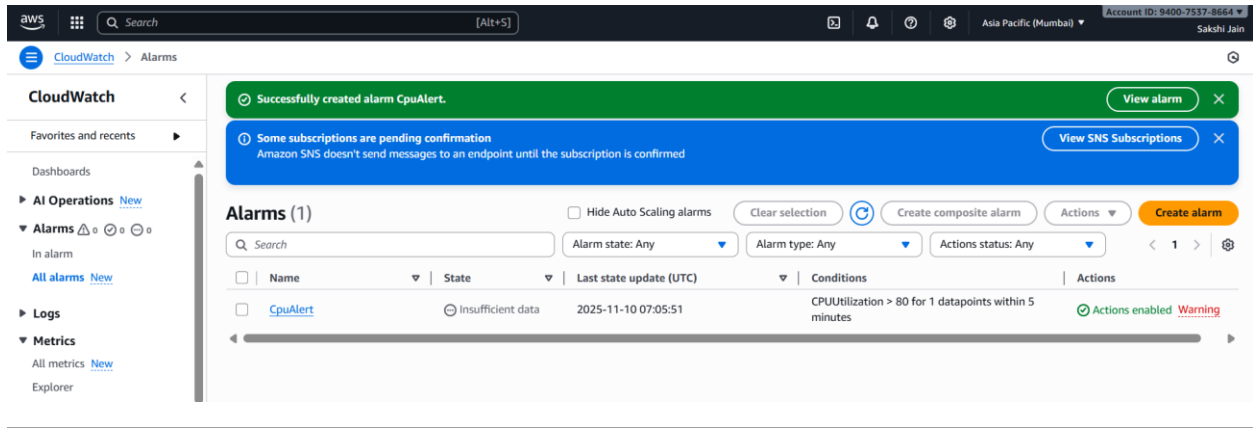
Alarm description - optional View formatting guidelines

Edit Preview

This is an H1
double asterisks will produce strong character
This is [an example](https://example.com/) inline link.

Up to 1024 characters (0/1024)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.



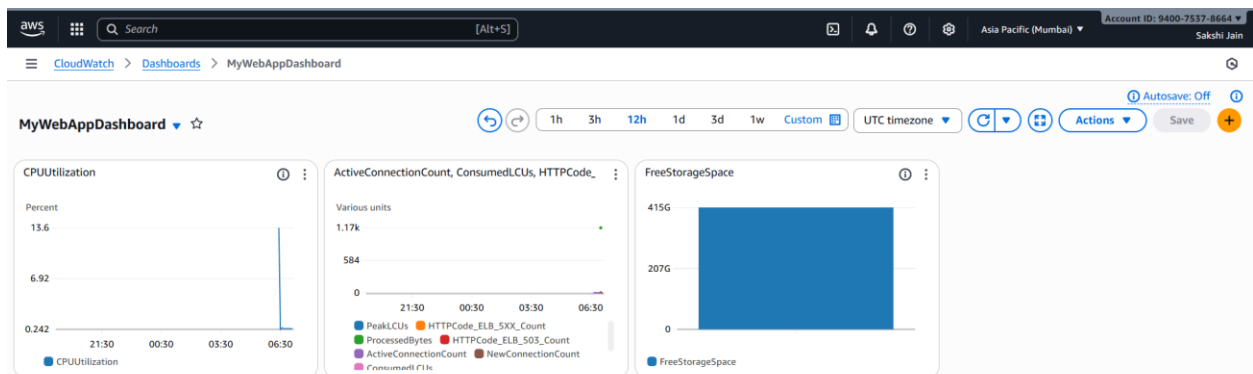
STEP 9: Create a CloudWatch Dashboard

Why:

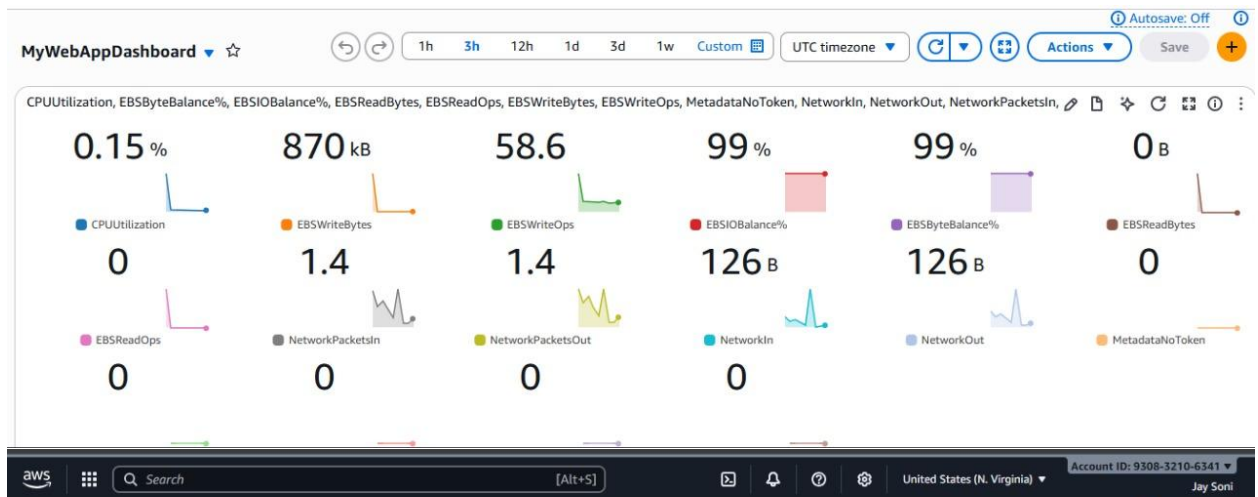
To see real-time data visually (CPU, network, number of instances, etc.)

How:

1. Go to **CloudWatch** → **Dashboards** → **Create Dashboard**
2. Name: MyWebAppDashboard
3. Add Widgets:
 - EC2 → Average CPU Utilization (Line graph)
 - ALB → Request Count (Line graph)
 - Auto Scaling → Group Desired Capacity (Number)
 - RDS → Free Storage Space (Bar chart)
4. Arrange and Save.



Autoscaling Load Balance

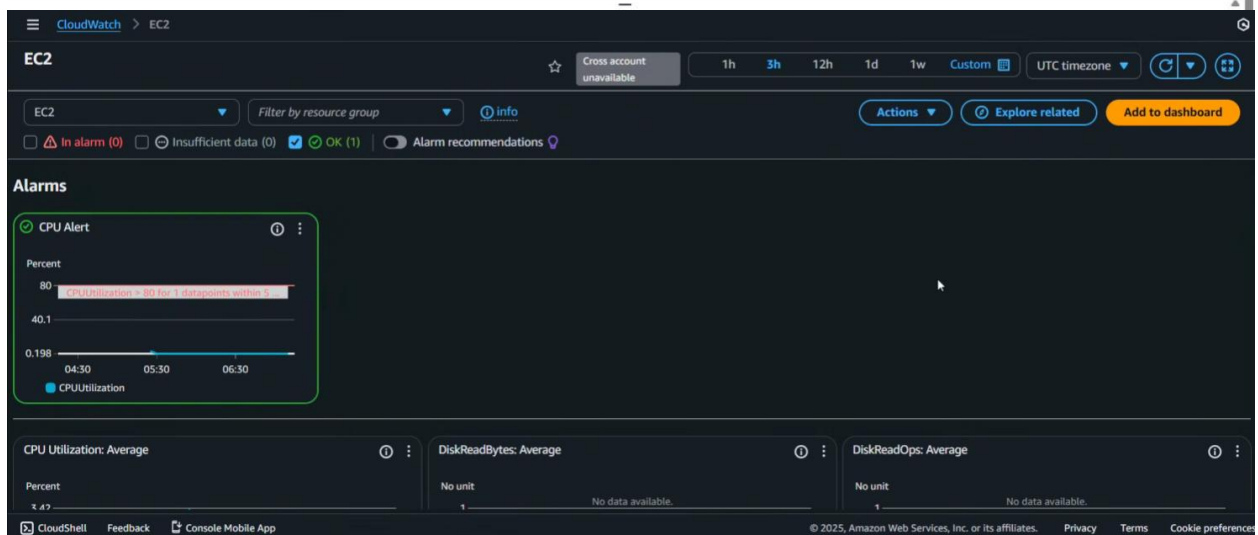


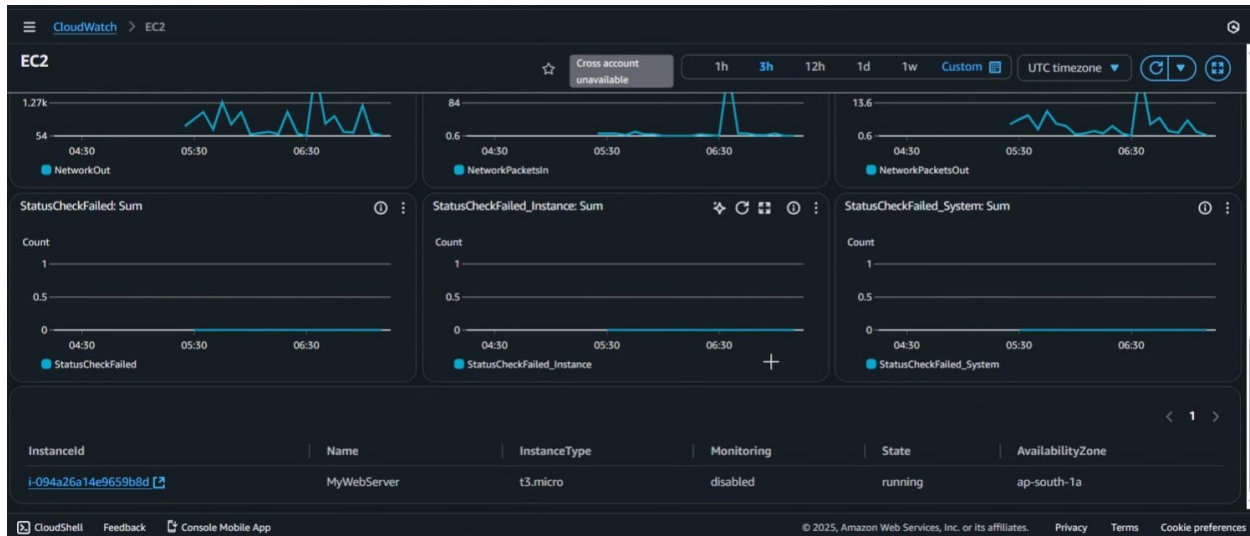
EC2 > Instances

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive) All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS
☒	MyWebServer	i-0d687fdae73f52445	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a	ec2-54-146-240-194.cr
☐		i-0d1267c2163d64be6	Running	c7i-flex.large	3/3 checks passed	View alarms +	us-east-1b	-





STEP 10: Test Everything Together

1. Visit the Load Balancer:

- Go to **EC2** → **Load Balancers** → **MyWebAppALB**
- Copy **DNS name**
- Paste in your browser → you should see your web app message.



2. Watch CloudWatch + SNS:

- You'll get an **email alert** when CPU > 80%
- Dashboard will show scaling in real-time

